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A	 PT. ABB Sakti Industri Jalan Gajah Tunggal km 1.0 Tangerang 15135, Indonesia								A																																										
B									B																																										
C	Project Name : MV Switchgear for MNA Serang Plant Switchgear Name : MVD-20 & MVD-21 20kV Boiler Plant Panel Type : UNISEC Customer Name : PT Multimas Nabati Asahan Contract No. : BE318042 TITLE : SLD & GA Drawing								C																																										
D									D																																										
E									E																																										
F	<table><tr><td></td><td></td><td></td><td></td><td>Date Prep.</td><td>15.11.2018</td><td>PROJECT TITLE</td><td rowspan="3"> ABB EPMV PT. ABB Sakti Industri </td><td rowspan="3">COVER SHEET SLD & GA Drawing MVD-20 & MVD-21 20kV Boiler Plant</td><td>Scale</td><td></td><td>= H00</td></tr><tr><td>1</td><td>Issued for Approval</td><td>18.12.2018</td><td>YR</td><td>Prepared</td><td>YR</td><td rowspan="2">MV Switchgear for MNA Serang Plant</td><td>Customer</td><td>PT Multimas Nabati Asahan</td><td>& GAD</td></tr><tr><td>0</td><td>First Issued</td><td>19.11.2018</td><td>YR</td><td>Checked</td><td>AW</td><td>Drawing Nbr.</td><td></td><td>Sh. 1</td></tr><tr><td>Ind.</td><td>Revision</td><td>Date</td><td>Name</td><td>Approved</td><td>AI</td><td>Based on:</td><td>BE318042</td><td>Customer Doc. Nbr.</td><td></td><td>Cont. 23</td></tr></table>												Date Prep.	15.11.2018	PROJECT TITLE	ABB EPMV PT. ABB Sakti Industri	COVER SHEET SLD & GA Drawing MVD-20 & MVD-21 20kV Boiler Plant	Scale		= H00	1	Issued for Approval	18.12.2018	YR	Prepared	YR	MV Switchgear for MNA Serang Plant	Customer	PT Multimas Nabati Asahan	& GAD	0	First Issued	19.11.2018	YR	Checked	AW	Drawing Nbr.		Sh. 1	Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042	Customer Doc. Nbr.		Cont. 23	F
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A	SHEET INDEX								A		
B	PAGE		PAGE NAME		REVISION						
	==MVD-20_MVD-21=H00&GAD/1		COVER SHEET		0	1					
	==MVD-20_MVD-21=H00&GAD/2		INDEX OF SHEETS		0	1					
	==MVD-20_MVD-21=H00&GAD/3		GENERAL CHARACTERISTICS		0	1					
	==MVD-20_MVD-21=H00&GAD/4		REFERENCE DESIGNATIONS		0	1					
	==MVD-20_MVD-21=H00&GAD/5		REFERENCE DESIGNATIONS		0	1					
	==MVD-20_MVD-21=H00&GAD/6		REFERENCE DESIGNATIONS		0	1					
	==MVD-20_MVD-21=H00&GAD/7		REFERENCE DESIGNATIONS		0	1					
	==MVD-20_MVD-21=H00&GAD/8		SINGLE LINE DIAGRAM		0	1					
	==MVD-20_MVD-21=H00&GAD/9		SINGLE LINE DIAGRAM		0	1					
	==MVD-20_MVD-21=H00&GAD/10		SINGLE LINE DIAGRAM		0	1					
	==MVD-20_MVD-21=H00&GAD/11		INSTALLATION RULES		0	1					
	==MVD-20_MVD-21=H00&GAD/12		FRONT VIEW & GA		0	1					
	==MVD-20_MVD-21=H00&GAD/13		FRONT VIEW & GA		0	1					
	==MVD-20_MVD-21=H00&GAD/14		FOUNDATION FRAMES		0	1					
	==MVD-20_MVD-21=H00&GAD/15		FOUNDATION FRAMES		0	1					
	==MVD-20_MVD-21=H00&GAD/16		FOUNDATION FRAMES DETAILS		0	1					
	==MVD-20_MVD-21=H00&GAD/17		FOUNDATION FRAMES DETAILS		0	1					
	==MVD-20_MVD-21=H00&GAD/18		FOUNDATION FRAMES DETAILS		0	1					
	==MVD-20_MVD-21=H00&GAD/19		SECTION VIEW		0	1					
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	==MVD-20_MVD-21=H00&GAD/21		SECTION VIEW		0	1					
	==MVD-20_MVD-21=H00&GAD/22		SECTION VIEW		0	1					
	==MVD-20_MVD-21=H00&GAD/23		SECTION VIEW		0	1					
D									D		
E									E		
F									F		
				Date Prep.	15.11.2018	PROJECT TITLE		ABB EPMV	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR	MV Switchgear for MNA Serang Plant			Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW			PT. ABB Sakti Industri	Drawing Nbr.		Sh. 2
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042			Customer Doc. Nbr.		Cont. 23
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A	CHARACTERISTICS (IN COMPLIANCE WITH STANDARD IEC 62271-200)				NOTES				A		
B	SWITCHGEAR VERSION			= UNISEC	COLOUR OF FRONT DOORS			= RAL 7035			
	RATED VOLTAGE (Un)			= 24 kV	SIDEWALLS PAINTED			= YES			
	OPERATING VOLTAGE			= 20 kV	SIDES			= L+R			
	RATED FREQUENCY (fr)			= 50 Hz							
C	RATED LIGHTNING IMPULSE WITHSTAND VOLTAGE (Up)			= 125 kV							
	RATED POWER FREQUENCY WITHSTAND VOLTAGE (Ud)			= 50 kV							
	RATED CURRENT OF MAIN BUSBARS (Ir)			= 630 A							
	RATED SHORT-TIME WITHSTAND CURRENT (Ik)			= 20 kA							
D	RATED PEAK WITHSTAND CURRENT (Ip)			= 50 kA							
	RATED DURATION OF SHORT CIRCUIT (tk)			= 1 s							
	INTERNAL ARC CLASSIFICATION (IAC)			= AFLR / 16kAx1s							
	AMBIENT CONDITION			= NORMAL							
E	AMBIENT AIR TEMPERATURE			= -5°C...+40°C							
	DEGREE OF PROTECTION (OPERATION SEATS EXCLUDED)			= IP3X							
	DEGREE OF PROTECTION WITH OPEN DOORS			= IP2X							
	RATED SUPPLY VOLTAGE OF CONTROL AND SIGNALLING CIRCUITS (Ua)			= 230VAC							
F	RATED SUPPLY VOLTAGE OF SPRING CHARGING MOTOR (Ua)			= 230VAC							
	RATED SUPPLY VOLTAGE OF LIGHTING AND HEATING CIRCUITS (Ua)			= 230VAC							
	TYPE OF CABLE FOR LOW VOLTAGE CONDUCTORS			= 450/750V							
	TYPE OF CABLE FOR LOW VOLTAGE CONDUCTORS			= PVC							
G	CROSS-SECTION OF CONDUCTORS FOR CURRENT CIRCUITS			= 2.5 mm²/BLACK							
	CROSS-SECTION OF CONDUCTORS FOR VOLTAGE CIRCUITS			= 1.5 mm²/BLACK							
	CROSS-SECTION OF CONDUCTORS FOR MOTORS			= 2.5 mm²/BLACK							
	CROSS-SECTION OF OTHER CONDUCTORS (CONTROL AND SIGNALLING CIRCUITS)			= 1.5 mm²/BLACK							
H	CROSS-SECTION OF CONDUCTORS FOR INTERCONNECTIONS			= 1.5 mm²/BLACK							
	CROSS-SECTION OF CONDUCTORS FOR INTERCONNECTIONS OF SUPPLY VOLTAGE			= 2.5 mm²/BLACK							
	COMMUNICATION CABLE			= N/A							
				IDENTIFICATION OF CONDUCTORS							
				FOR IDENTIFICATION OF CONDUCTORS THE "LOCAL END CONNECTION LABELLING" SYSTEM IS USED, IN COMPLIANCE WITH STANDARD IEC 62491 PARAGRAPH 6.2. THIS SYSTEM FORESEES THAT THE MARKING OF A CONDUCTOR END SHOWS THE DESIGNATION OF THE TERMINAL TO WHICH IT IS CONNECTED. IN COMPLIANCE WITH STANDARD IEC 61666 EACH CONDUCTOR IS IDENTIFIED WITH THE REFERENCE DESIGNATION OF THE CONNECTED ELECTRICAL COMPONENT FOLLOWED BY SIGN ":" (COLON), FOLLOWED BY THE TERMINAL DESIGNATION. FOR EXAMPLE THE CONDUCTOR CONNECTED TO TERMINAL "10" OF THE COMPONENT WITH DESIGNATION "-XA" HAS THE MARKING "-XA:10". THE CONDUCTOR MARKING IS PLACED IN SUCH A WAY TO READ IT ALWAYS FROM THE TERMINAL TOWARDS THE CONDUCTOR, AS SHOWN BELOW							
				01:VQX- 9 10 11 -XDA:10							

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A	REFERENCE DESIGNATION OF OBJECTS IN ELECTRICAL DOCUMENTS														A
	(IN COMPLIANCE WITH STANDARD IEC 81346-2 AND ABB TECHNICAL STANDARD 2NBA000001)														
B	DESIGNATION	DESCRIPTION													B
	-AA	MULTIFUNCTION UNIT (CENTRAL UNIT)													
	-BAD1,-BAD4	CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L1													
	-BAD2,-BAD5	CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L2													
	-BAD3,-BAD6	CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L3													
	-BAR	VOLTAGE PROTECTION RELAY													
	-BAS1	VOLTAGE SENSOR LOCATED ON PHASE L1													
	-BAS2	VOLTAGE SENSOR LOCATED ON PHASE L2													
	-BAS3	VOLTAGE SENSOR LOCATED ON PHASE L3													
	-BAT1	VOLTAGE TRANSFORMER LOCATED ON PHASE L1													
C	-BAT2	VOLTAGE TRANSFORMER LOCATED ON PHASE L2													C
	-BAT3	VOLTAGE TRANSFORMER LOCATED ON PHASE L3													
	-BAT4	VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L1													
	-BAT5	VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L2													
	-BAT6	VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L3													
	-BCD	DIFFERENTIAL PROTECTIVE RELAY													
	-BCF	FEEDER PROTECTION RELAY													
	-BCG	GENERATOR PROTECTION RELAY													
	-BCM	MOTOR PROTECTION RELAY													
	-BCN	NEUTRAL (RESIDUAL) CURRENT TRANSFORMER													
D	-BCP	TRANSFORMER PROTECTION RELAY													D
	-BCR	CURRENT PROTECTION RELAY													
	-BCS1	CURRENT SENSOR LOCATED ON PHASE L1													
	-BCS2	CURRENT SENSOR LOCATED ON PHASE L2													
	-BCS3	CURRENT SENSOR LOCATED ON PHASE L3													
	-BCT1	CURRENT TRANSFORMER LOCATED ON PHASE L1													
	-BCT2	CURRENT TRANSFORMER LOCATED ON PHASE L2													
	-BCT3	CURRENT TRANSFORMER LOCATED ON PHASE L3													
	-BCT4,-BCT7	CURRENT TRANSFORMER LOCATED ON PHASE L1													
	-BCT5,-BCT8	CURRENT TRANSFORMER LOCATED ON PHASE L2													
E	-BCT6,-BCT9	CURRENT TRANSFORMER LOCATED ON PHASE L3													E
	-BCZ	DISTANCE PROTECTION RELAY													
	-BEF	FREQUENCY PROTECTION RELAY													
	-BER	SUPERVISION RELAYS													
	-BES	SYNCHRONIZING RELAY													
	-BET	THERMAL PROTECTION RELAY													
	-BGA1	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CIRCUIT-BREAKER COMPARTMENT													
	-BGA2	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN BUSBAR COMPARTMENT (SINGLE BUSBAR SYSTEM) OR IN FRONT BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM)													
	-BGA3	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CABLE COMPARTMENT													
	-BGA4	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN REAR BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM)													
F	-BGB1...-BGB3	POSITION SWITCHES OF CIRCUIT-BREAKER													F
	-BGB4	POSITION SWITCH OF CIRCUIT-BREAKER: PASSING CONTACT CLOSING MOMENTARILY WHEN THE BREAKER OPENS													

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A	-BGI31		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION		-CA		CAPACITORS		A						
	-BGI32		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION		-CA1		CAPACITOR FOR FRONT VENTILATOR								
	-BGI33		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)		-CA2		CAPACITOR FOR REAR VENTILATOR								
	-BGI41...		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION		-EA1		LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.								
	-BGI42		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION		-EA2		LIGHTING LAMP LOCATED IN CABLE COMPARTMENT								
B	-BGI43...		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION		-EA3		LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE		B						
	-BGI51		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION		-EA4		LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT								
	-BGI52		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION		-EB1		HEATER LOCATED IN CABLE COMPARTMENT								
	-BGI53		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION		-EB2		HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT								
	-BGK		POSITION SWITCH OF KEY LOCK		-EB3		HEATER LOCATED IN MOTOR CABINET								
C	-BGL1		POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1		-EB4		HEATER FOR VENTILATION TEST		C						
	-BGS1....-BGS4		POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS		-EB5		HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.								
	-BGS5		PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION		-EB7		HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE								
	-BGT1		POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION		-EB8		HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE								
	-BGT2		POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION		-FA1		SURGE ARRESTER LOCATED ON PHASE L1								
D	-BGT3		POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION		-FA2		SURGE ARRESTER LOCATED ON PHASE L2		D						
	-BGT4		POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION		-FA3		SURGE ARRESTER LOCATED ON PHASE L3								
	-BGT5		POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION		-FCD		FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS								
	-BGT6		POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION		-FCF1		MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1								
	-BGT7		POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION		-FCF2		MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2								
E	-BGT8		POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION		-FCF3		MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3		E						
	-BGT11		PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION		-FCM1		MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER								
	-BGT12		PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION		-FCM2...-FCM10		MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS								
	-BGV1		POSITION SWITCH OF FRONT VENTILATOR		-FCM11		MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1								
	-BGV2		POSITION SWITCH OF REAR VENTILATOR		-FCM12		MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2								
F	-BM		HYGROSTAT		-FCM13		MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3		F						
	-BPA4		PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING		-FCT		THERMAL OVERLOAD RELEASES								
	-BPA5		PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING		-GA		GENERATORS								
	-BPA6		PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING		-GQ1		FRONT VENTILATOR								
	-BPA7		PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING		-GQ2		REAR VENTILATOR								
-BPS1		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER		-GQ3		VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK		F							
-BPS2		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2		-KFA1		AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE									
-BPS3		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3		-KFA2		AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE									
-BR		FLAME DETECTORS, SMOKE DETECTORS		-KFA3....-KFA9		AUXILIARY RELAYS OR CONTACTORS									
-BT		THERMOSTAT		-KFC		CLOSING RELAYS OR CONTACTORS									
-BT1		THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS		-KFI		INTEGRATED CIRCUITS		F							
-BT2		THERMOSTAT FOR ENERGIZATION OF VENTILATORS		-KFL		LOCKOUT RELAY									
-BT3		THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM		-KFM		MICROPROCESSORS									
-BU		BUCHHOLZ RELAY		-KFN		ANTIPUMPING RELAYS									
-BX1		UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING		-KFO		OPENING RELAYS OR CONTACTORS									
-BX2		ADDITIONAL CURRENT SENSING UNIT FOR DETECTION OF INTERNAL ARCING		-KFP		PROGRAMMABLE LOGIC CONTROLLERS (PLC)		F							
-BZ,-BZ4		MULTIFUNCTION SENSORS		-KFR		RECLOSING RELAYS									
-BZ1		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L1		-KFS		SYNCHRONIZING DEVICES									
-BZ2		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L2		-KFT		AUXILIARY TIME RELAYS, DELAY ELEMENTS									
-BZ3		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L3		-KFZ		CONTROLLERS									
-MAD		MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD		-MBC		CLOSING RELEASE OF CIRCUIT-BREAKER		F							
-MAD1		MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1		-MAD		MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD									
-MAD2		MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2		-MAD1		MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1									
-MAD9		MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS		-MAD2		MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2									
				-MAD9		MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS									
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A	-MAE	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE						-QBH	MANUAL CIRCUIT-BREAKERS					
	-MAE1	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2						-QBL	LINKS					
	-MAE8	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS						-QBM	MINIATURE SWITCH-DISCONNECTORS					
	-MAS	MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING						-QBS	SWITCH-DISCONNECTORS					
	-MAT	MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT						-QBT	TRUCK					
	-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER						-QCA	EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING					
	-MBC4	CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS						-QCE	EARTHING SWITCH					
	-MBO1	FIRST OPENING RELEASE OF CIRCUIT-BREAKER						-QCE1	EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)					
	-MBO2	SECOND OPENING RELEASE OF CIRCUIT-BREAKER						-QCE2	EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)					
	-MBO3	OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER						-QQ	CLUTCHES					
B	-MBO4	OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS						-RAA	ANTIFERRORESONANCE RESISTOR					
	-MBU	UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER						-RAD	DIODES					
	-PFB	BLUE SIGNAL LAMPS						-RAI	INDUCTORS					
	-PFF	FLAG RELAYS						-RAR	RESISTORS					
	-PFG	GREEN SIGNAL LAMPS						-RF	FILTERS					
C	-PFR	RED SIGNAL LAMPS						-RLE1	ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING					
	-PFS	SHORT CIRCUIT INDICATORS						-RLE2	ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT					
	-PFV	VOLTAGE INDICATORS						-RLE3	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE					
	-PFV1	VOLTAGE INDICATORS ON FEEDER SIDE						-RLE4	ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION					
	-PFV2	VOLTAGE INDICATORS ON BUSBAR SIDE						-RLE5	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD					
	-PFW	WHITE SIGNAL LAMPS						-RLE6	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1					
	-PFX	CROSS INDICATORS, ELECTROMECHANICAL INDICATORS						-RLE7	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2					
	-PFY	YELLOW SIGNAL LAMPS						-RLE8	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2					
	-PGA	AMMETERS						-RLE9	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS					
	-PGC	COUNTERS						-RLE11	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4					
D	-PGF	FREQUENCYMETERS						-RLE12	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5					
	-PGH	HOURMETERS						-SFA	AMMETRIC SWITCHES					
	-PGI	PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE						-SFC	CONTROL SWITCHES, CLOSING PUSH-BUTTONS					
	-PGI1	HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)						-SFL	LOCKING CONTACTS					
	-PGI2	HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)						-SFO	OPENING PUSH-BUTTONS					
	-PGJ	ACTIVE ENERGY METERS						-SFR	RESET PUSH-BUTTONS					
	-PGK	REACTIVE ENERGY METERS						-SFS	SELECTOR SWITCHES					
	-PGM	MULTIFUNCTION INDICATORS						-SFT	TEST PUSH-BUTTONS					
	-PGP	POWER-FACTOR METERS						-SFU	UNLOCKING PUSH-BUTTONS					
	-PGQ	VARMETERS						-SFV	VOLTMETRIC SWITCHES					
E	-PGS	SYNCHRONOSCOPES						-TA	POWER TRANSFORMERS					
	-PGV	VOLTMETERS						-TB	CONVERTER					
	-PGW	WATTMETERS						-TB1	RECTIFIER FOR FIRST OPENING RELEASE					
	-PJ	ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)						-TB2	RECTIFIER FOR SECOND OPENING RELEASE					
	-QAB	CIRCUIT-BREAKERS						-TB3	RECTIFIER FOR CLOSING RELEASE					
	-QAC	CONTACTORS (FOR POWER)						-TB4	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING					
	-QBD	DISCONNECTORS						-TB5	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT					
	-QBD1	FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB6	RECTIFIER FOR UNDERVOLTAGE RELEASE					
	-QBD2	REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB7	ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER					
	-QBD3	FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)												
F	-QBD4	TWO-WAY DISCONNECTOR												
	-QBD5	TWO-WAY DISCONNECTOR ON BUSBAR RISER												

				Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	REFERENCE DESIGNATIONS		Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR	MV Switchgear for MNA Serang Plant				SLD & GA Drawing		Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW			PT. ABB Sakti Industri		MVD-20 & MVD-21 20kV Boiler Plant		Drawing Nbr.		Sh. 6
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042						Customer Doc. Nbr.		Cont. 23

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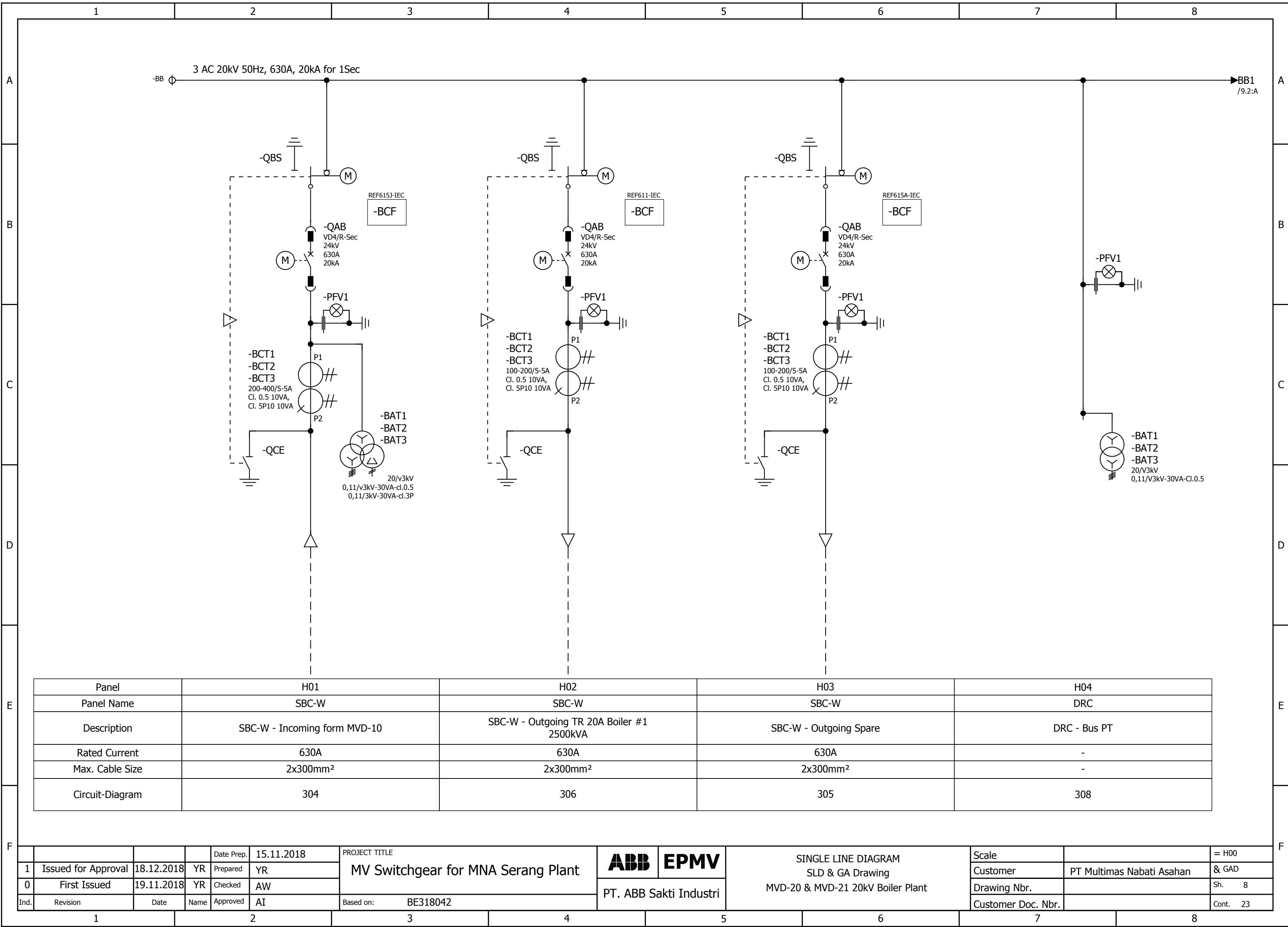
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A	-TFA		ACTIVE POWER TRANSDUCERS		-XD11	TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS		A
	-TFC		CURRENT TRANSDUCERS			TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING		
B	-TFF		FREQUENCY TRANSDUCERS		-XD13	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION		B
	-TFJ		ACTIVE ENERGY TRANSDUCERS			TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS		
C	-TFK		REACTIVE ENERGY TRANSDUCERS		-XD16	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS		C
	-TFM		MULTIFUNCTION TRANSDUCERS			TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS		
D	-TFP		POWER-FACTOR TRANSDUCERS		-XD17	TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS		D
	-TFQ		REACTIVE POWER TRANSDUCERS			TERMINAL BLOCK FOR SPECIAL USE		
E	-TFS		SIGNAL CONVERTERS		-XDJ	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD		E
	-TFV		VOLTAGE TRANSDUCERS			TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS		
F	-WA		BUSBARS		-XDM	SEALABLE TERMINAL BLOCK FOR MEASUREMENT		F
	-WBC		POWER CABLES			CURRENT SOCKETS		
G	-WE		EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.		-XDS	TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB		G
	-WGC		CONTROL CABLES			CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK		
H	-WGS		COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)		-XDU1	TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK		H
	-WH		OPTICAL FIBRES			TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS		
I	-XDA		TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS		-XDV1	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1		I
	-XDA1		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1			CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2		
J	-XDA2		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2		-XDV2	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2		J
	-XDA3		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3			CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3		
K	-XDA5		CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER		-XDV3	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3		K
	-XDB		CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS			CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR		
L	-XDB1		CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)		-XDV4	CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR		L
	-XDB2		CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER			SUPPORT TERMINAL BLOCK		
M	-XDB10....-XDB89		CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS		-XDX	SUPPORT TERMINAL BLOCK		M
	-XDB7		CONNECTOR FOR TRUCK POSITION CONTACTS			EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE		
N	-XDB91		CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR		-XE1	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE		N
	-XDB92		CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR			EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE		
O	-XDB93		CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR		-XE2	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE		O
	-XDB95		CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET			EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE		
P	-XDC		CUSTOMER TERMINAL BLOCK		-XE5	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE		P
	-XDC1		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER			EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE		
Q	-XDC2		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR		-XE6	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE		Q
	-XDC3		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT			EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE		
R	-XDC4		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS		-XE7	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE		R
	-XDC5		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER			EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE		
S	-XDE		TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE		-XE8	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE		S
	-XDE1		TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE			EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT		
T	-XDE2		TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2		-XE9	EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT		T
	-XDE3		TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2					
U	-XDF		TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS					
	-XDF1		CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR					
V	-XDF2		CONNECTOR FOR REAR VENTILATOR CIRCUITS					
	-XDF3		CONNECTOR FOR FRONT VENTILATOR CIRCUITS					
W	-XDH		TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)					
	-XDI		TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)					

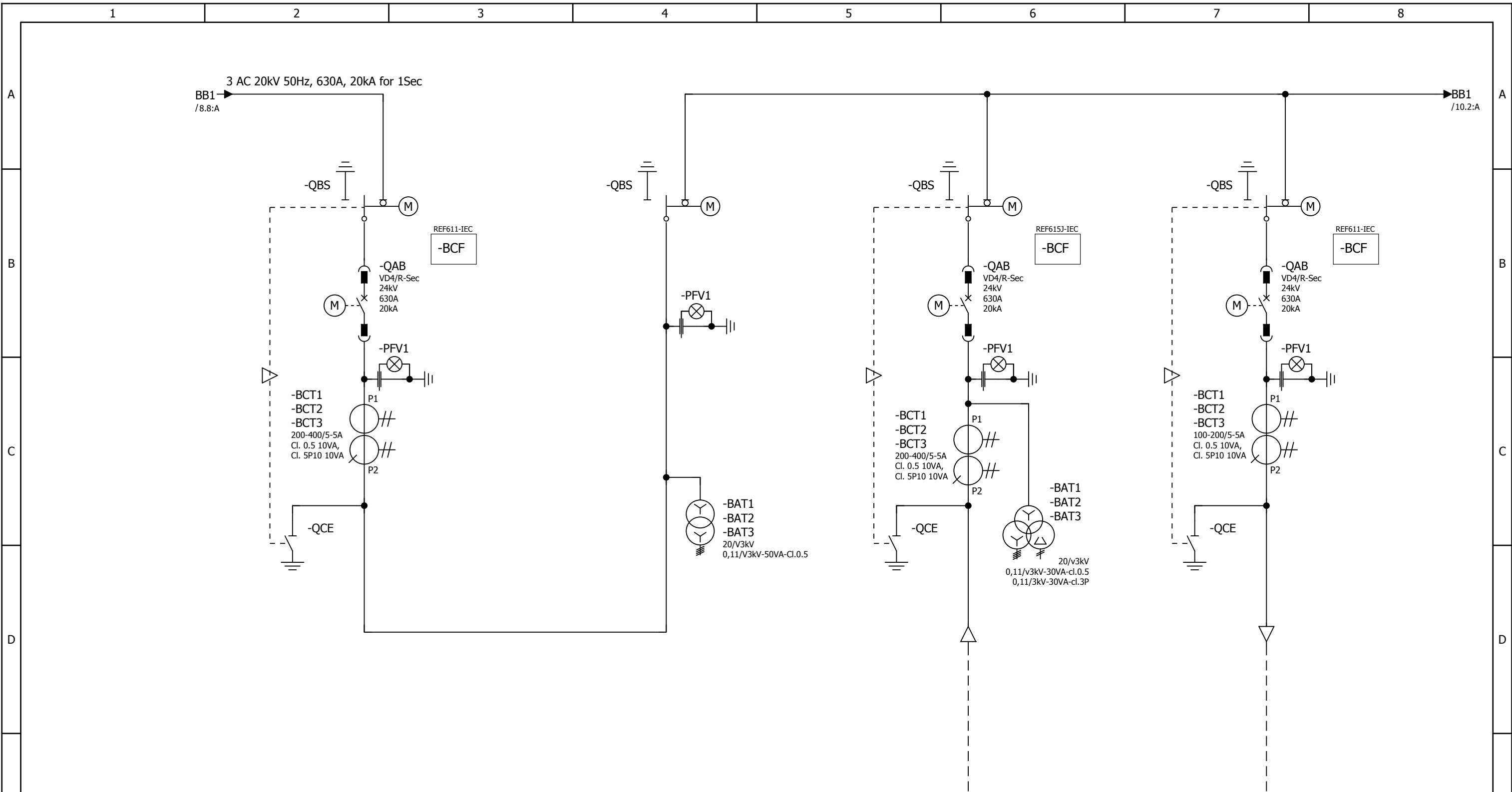
				Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	REFERENCE DESIGNATIONS SLD & GA Drawing MVD-20 & MVD-21 20kV Boiler Plant	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR	MV Switchgear for MNA Serang Plant					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW			PT. ABB Sakti Industri		Drawing Nbr.		Sh. 7	
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042				Customer Doc. Nbr.		Cont. 23	

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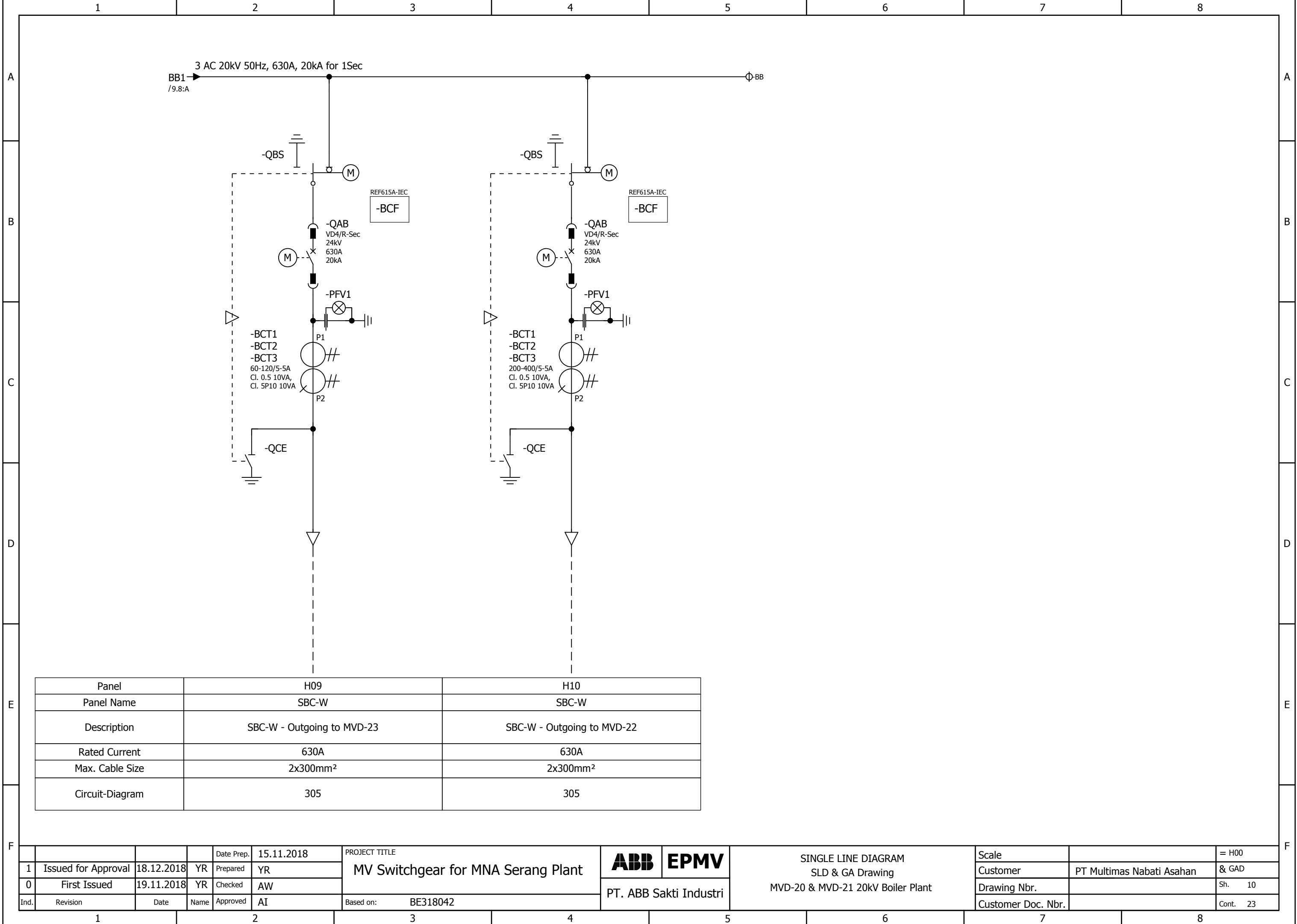


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Panel	H05	H06	H07	H08
Panel Name	SBS-W	SDS	SBC-W	SBC-W
Description	SBS-W - Bus Tie	SDS - Riser with VT	SBC-W - Incoming from MVD-11	SBC-W - Outgoing TR 20A Boiler #2 2500kVA
Rated Current	630A	-	630A	630A
Max. Cable Size	-	-	2x300mm ²	2x300mm ²
Circuit-Diagram	307	310	304	306

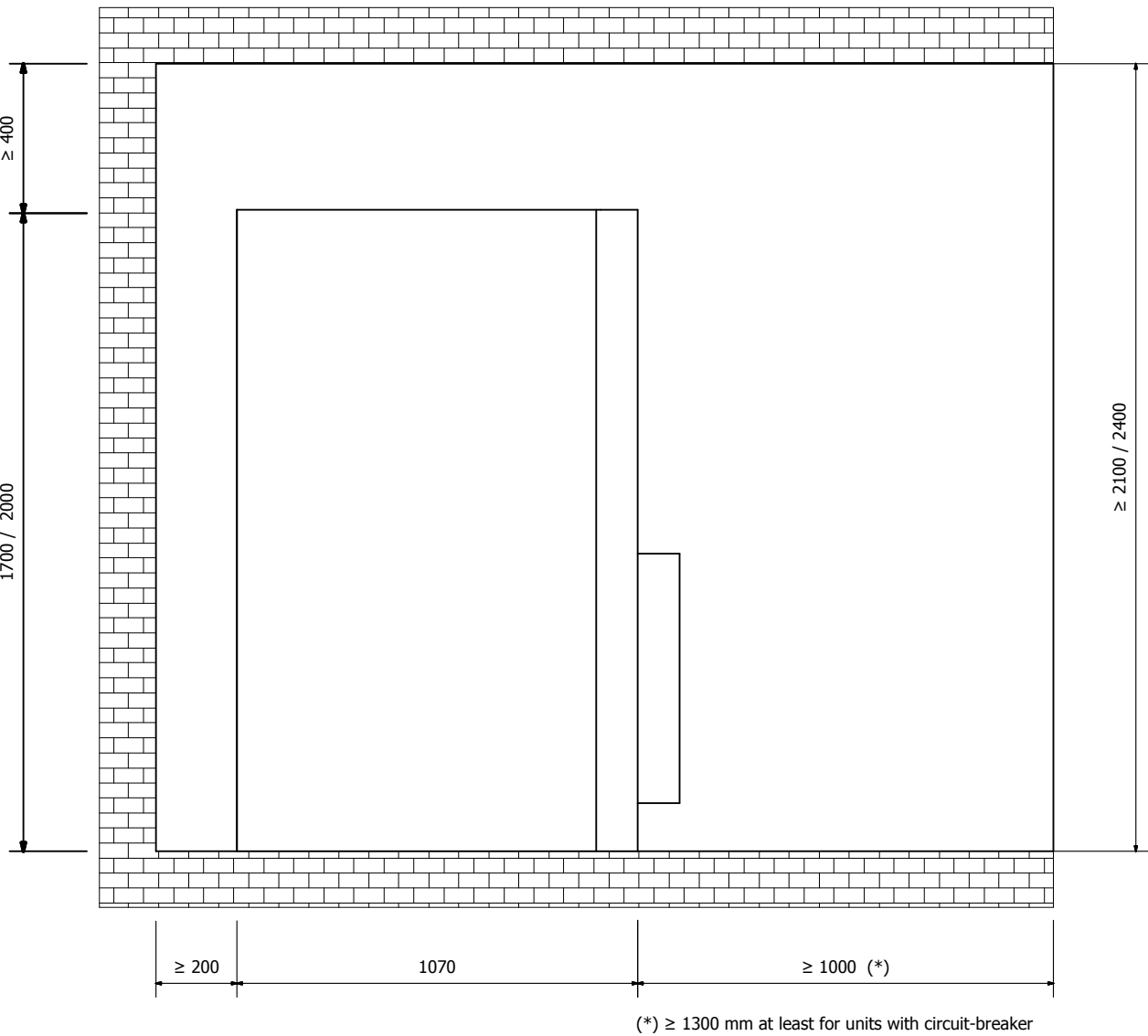
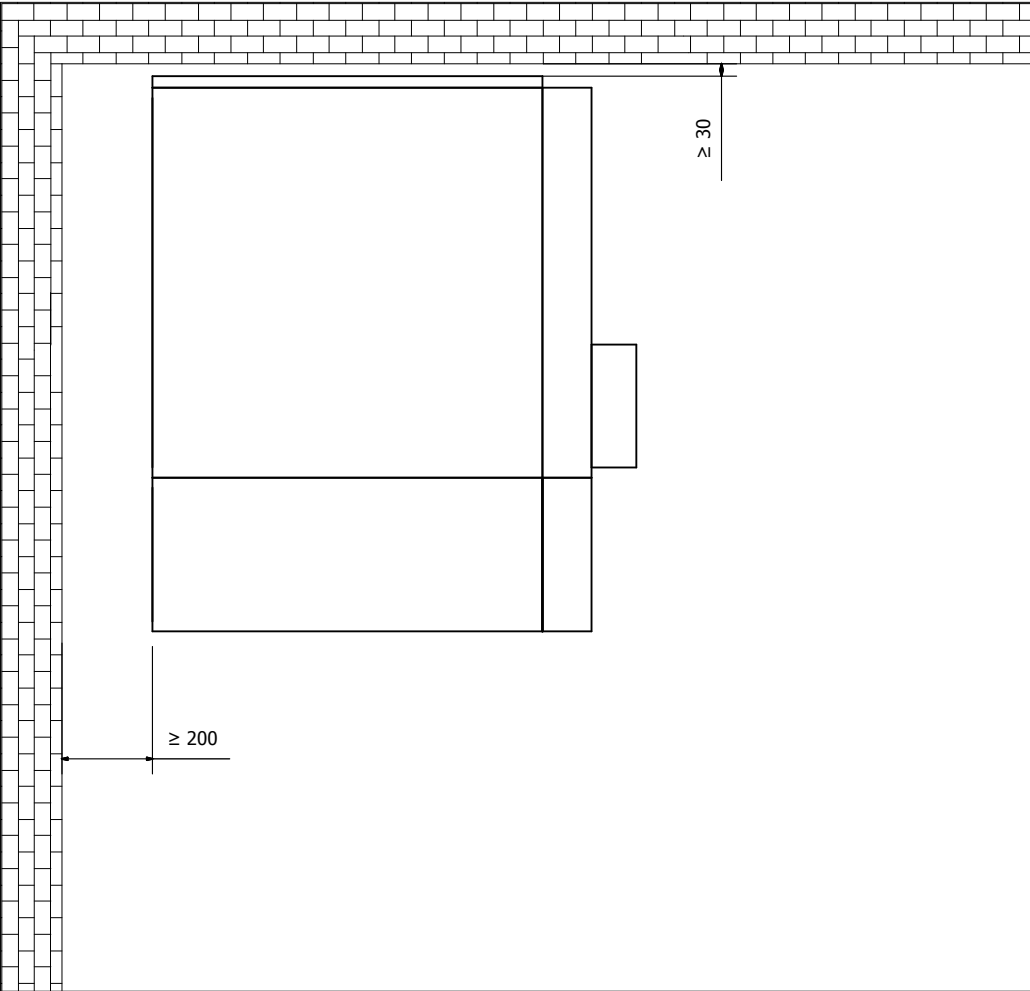
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Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042	PROJECT TITLE		ABB EPMV		SINGLE LINE DIAGRAM SLD & GA Drawing MVD-20 & MVD-21 20kV Boiler Plant		Scale	= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR	MV Switchgear for MNA Serang Plant		PT. ABB Sakti Industri						Customer	PT Multimas Nabati Asahan
0	First Issued	19.11.2018	YR	Checked	AW									Drawing Nbr.	Sh. 9
														Customer Doc. Nbr.	Cont. 23

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RULES FOR INSTALLATION IN THE ROOM

MINIMUM DISTANCES TO BE RESPECTED DURING INSTALLATION



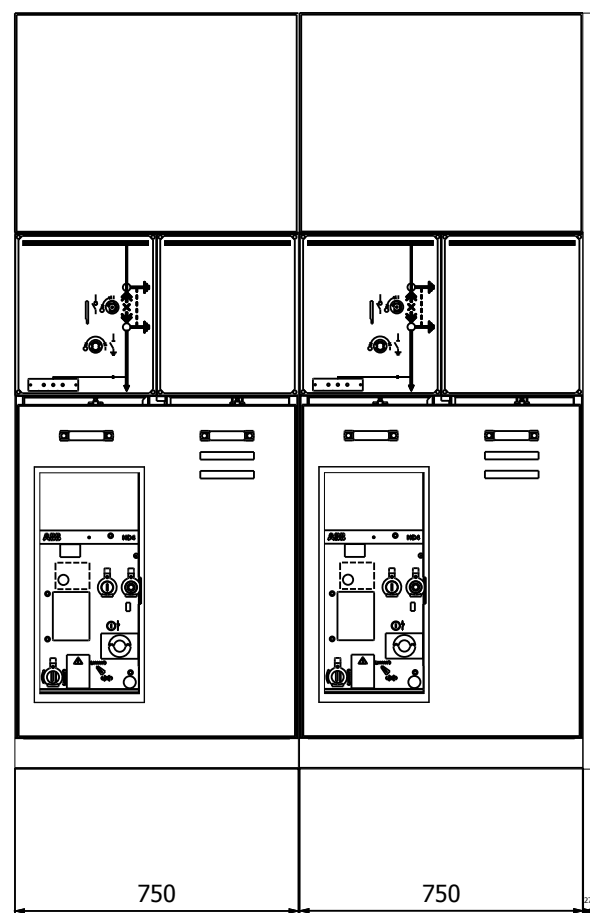
- IAC A-F SOLUTION UP TO 16kA 1sec
- ATTENTION : NO ACCESS TO REAR AND LATERAL OF THE SWG ROOM WHILE SWG IS IN SERVICE

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	INSTALLATION RULES SLD & GA Drawing MVD-20 & MVD-21 20kV Boiler Plant	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					PT. ABB Sakti Industri	Customer	PT Multimas Nabati Asahan
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.				Sh. 11	
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	Customer Doc. Nbr.			Cont. 23		

Panel	H01	H02	H03	H04	H05	H06	H07	H08
Panel Type	SBC-W	SBC-W	SBC-W	DRC	SBS-W	SDS	SBC-W	SBC-W
Description	SBC-W - Incoming	SBC-W - Outgoing TR 2500kVA	SBC-W - Spare Outgoing	DRC - Bus PT	SBS-W - Bus Tie	SDS - Riser with VT	SBC-W - Incoming	SBC-W - Outgoing TR 2500kVA

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	FRONT VIEW & GA SLD & GA Drawing MVD-20 & MVD-21 20kV Boiler Plant	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					PT. ABB Sakti Industri	Customer	PT Multimas Nabati Asahan
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.				Sh. 12	
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	Customer Doc. Nbr.			Cont. 23		

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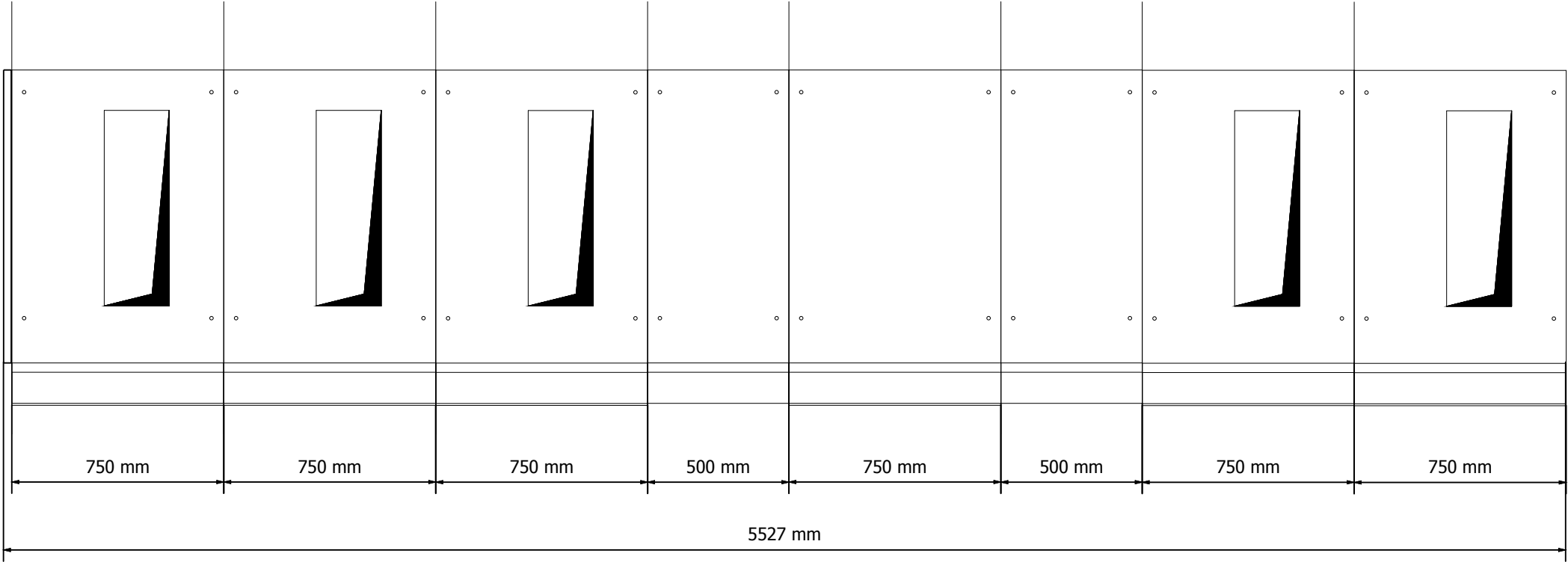


Panel	H09	H10
Panel Type	SBC-W	SBC-W
Description	SBC-W - Outgoing	SBC-W - Outgoing

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant			FRONT VIEW & GA SLD & GA Drawing MVD-20 & MVD-21 20kV Boiler Plant	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.			Sh. 13		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri	Customer Doc. Nbr.			Cont. 23	

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FOUNDATION FRAMES

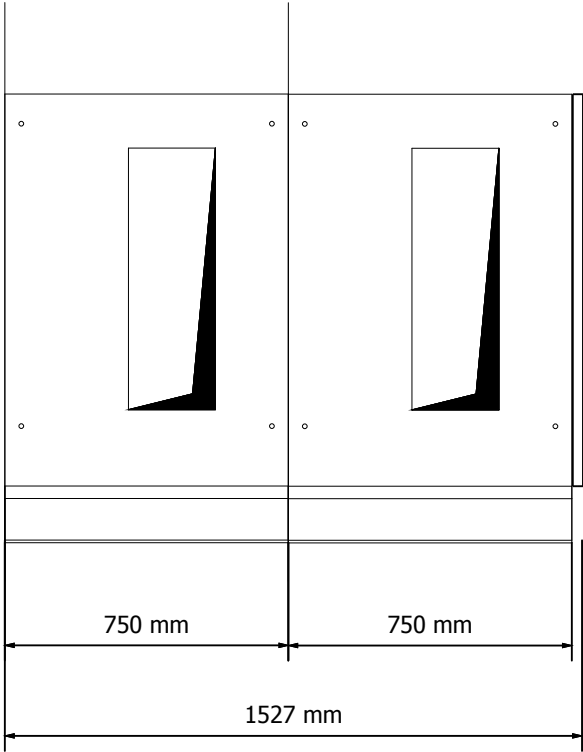


Panel	H01	H02	H03	H04	H05	H06	H07	H08
Panel Type	SBC-W	SBC-W	SBC-W	DRC	SBS-W	SDS	SBC-W	SBC-W
Description	SBC-W - Incoming	SBC-W - Outgoing TR 2500kVA	SBC-W - Spare Outgoing	DRC - Bus PT	SBS-W - Bus Tie	SDS - Riser with VT	SBC-W - Incoming	SBC-W - Outgoing TR 2500kVA

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	FOUNDATION FRAMES SLD & GA Drawing MVD-20 & MVD-21 20kV Boiler Plant	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.			Sh. 14		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 23

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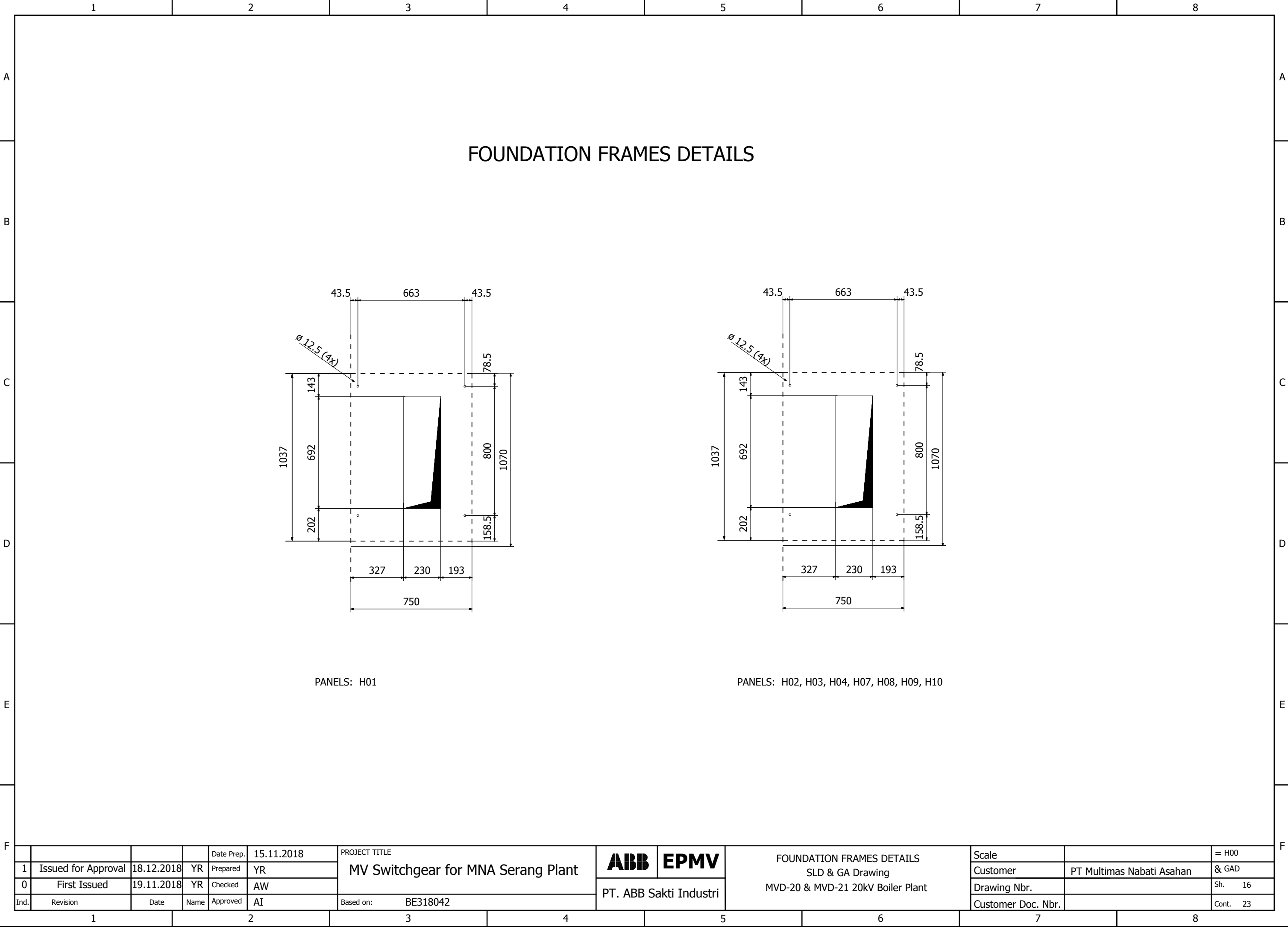
FOUNDATION FRAMES



Panel	H09	H10
Panel Type	SBC-W	SBC-W
Description	SBC-W - Outgoing	SBC-W - Outgoing

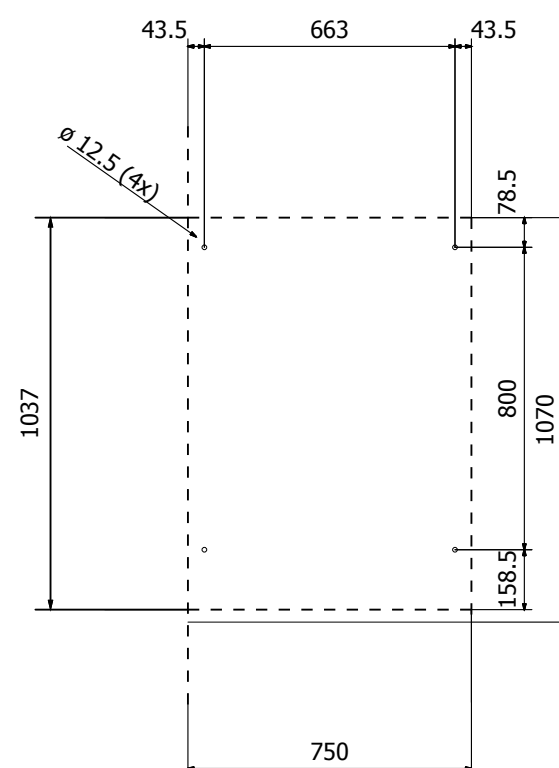
				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	FOUNDATION FRAMES SLD & GA Drawing MVD-20 & MVD-21 20kV Boiler Plant	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.			Sh. 15		
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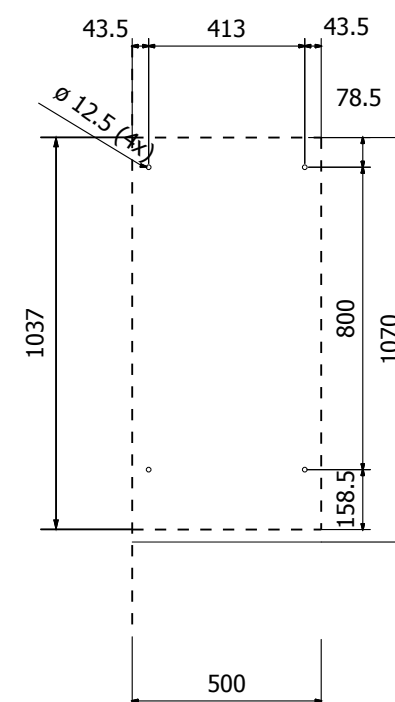


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FOUNDATION FRAMES DETAILS



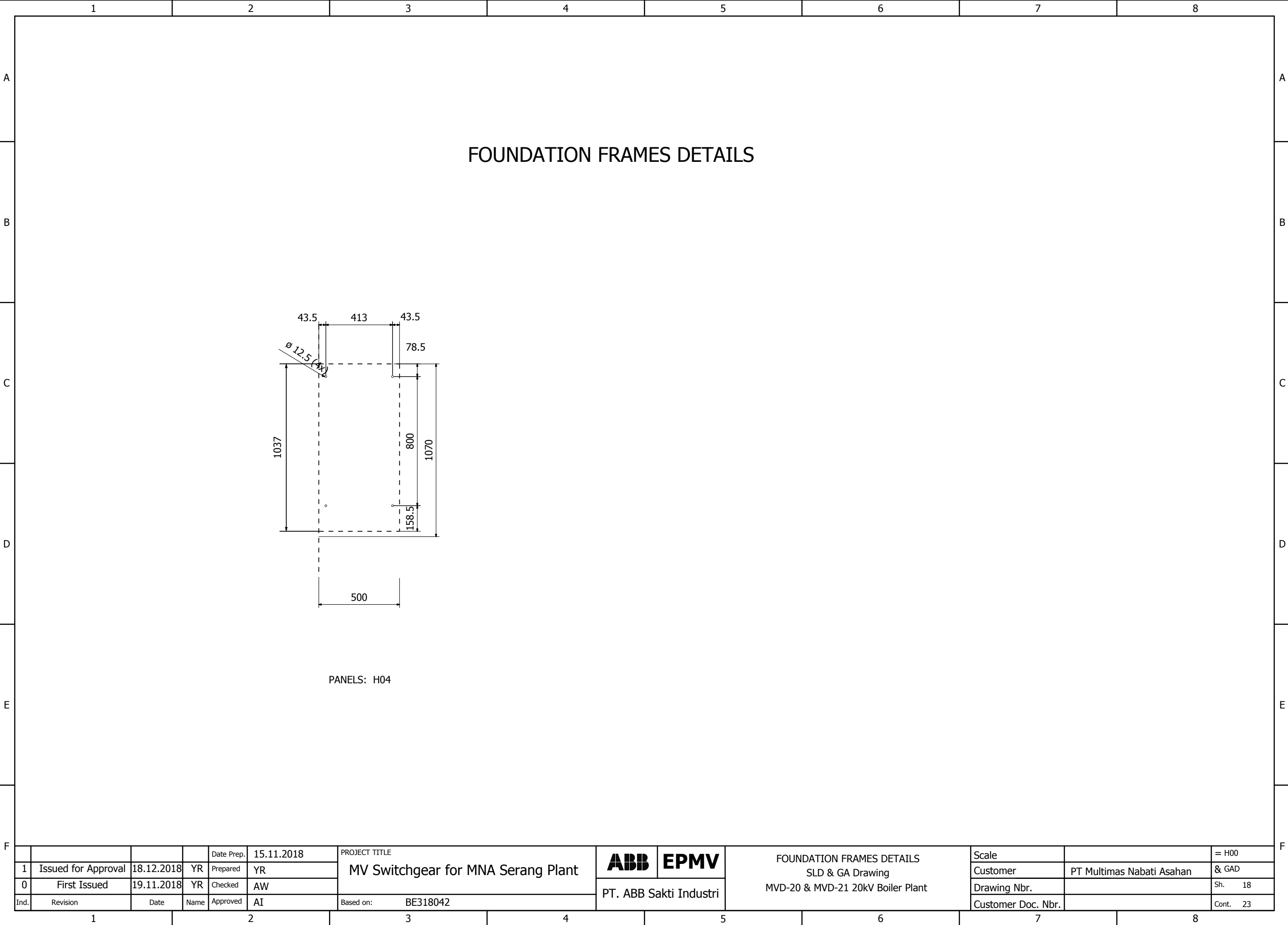
PANELS: H05



PANELS: H06

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	FOUNDATION FRAMES DETAILS SLD & GA Drawing MVD-20 & MVD-21 20kV Boiler Plant	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					PT. ABB Sakti Industri	Customer	PT Multimas Nabati Asahan
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.				Sh. 17	
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042	Customer Doc. Nbr.				Cont. 23

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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	FOUNDATION FRAMES DETAILS SLD & GA Drawing MVD-20 & MVD-21 20kV Boiler Plant	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.			Sh. 18		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 23

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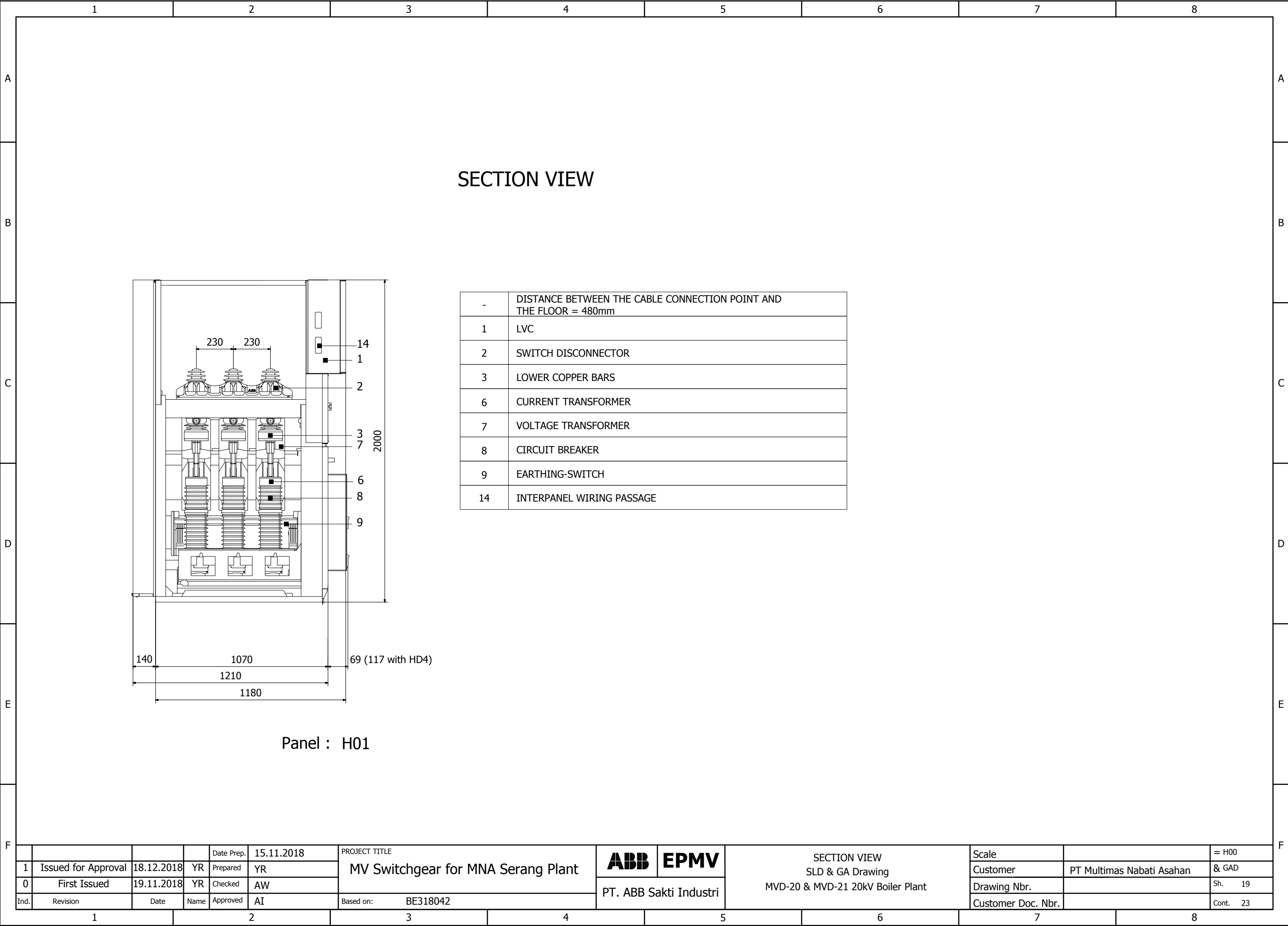
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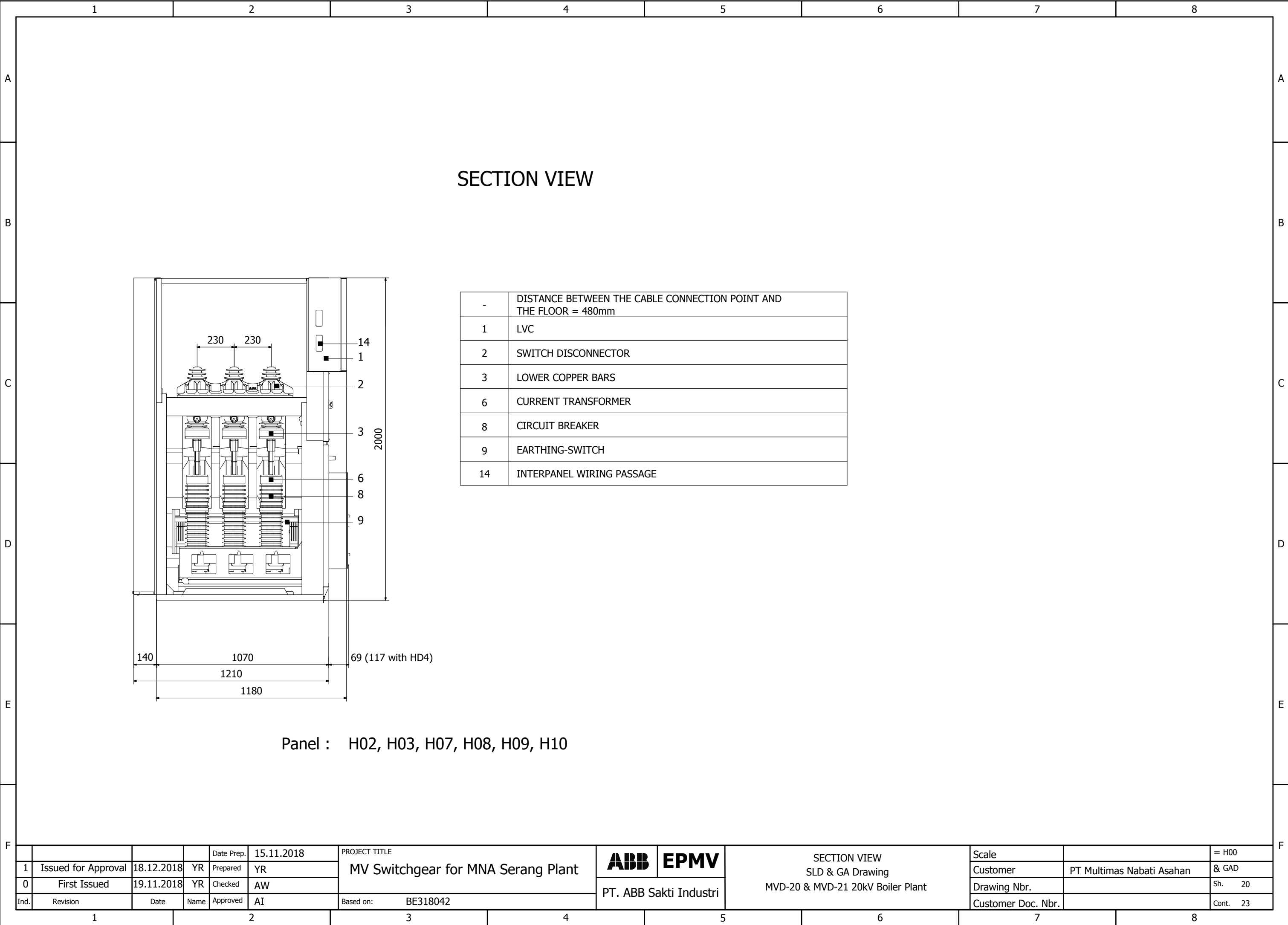
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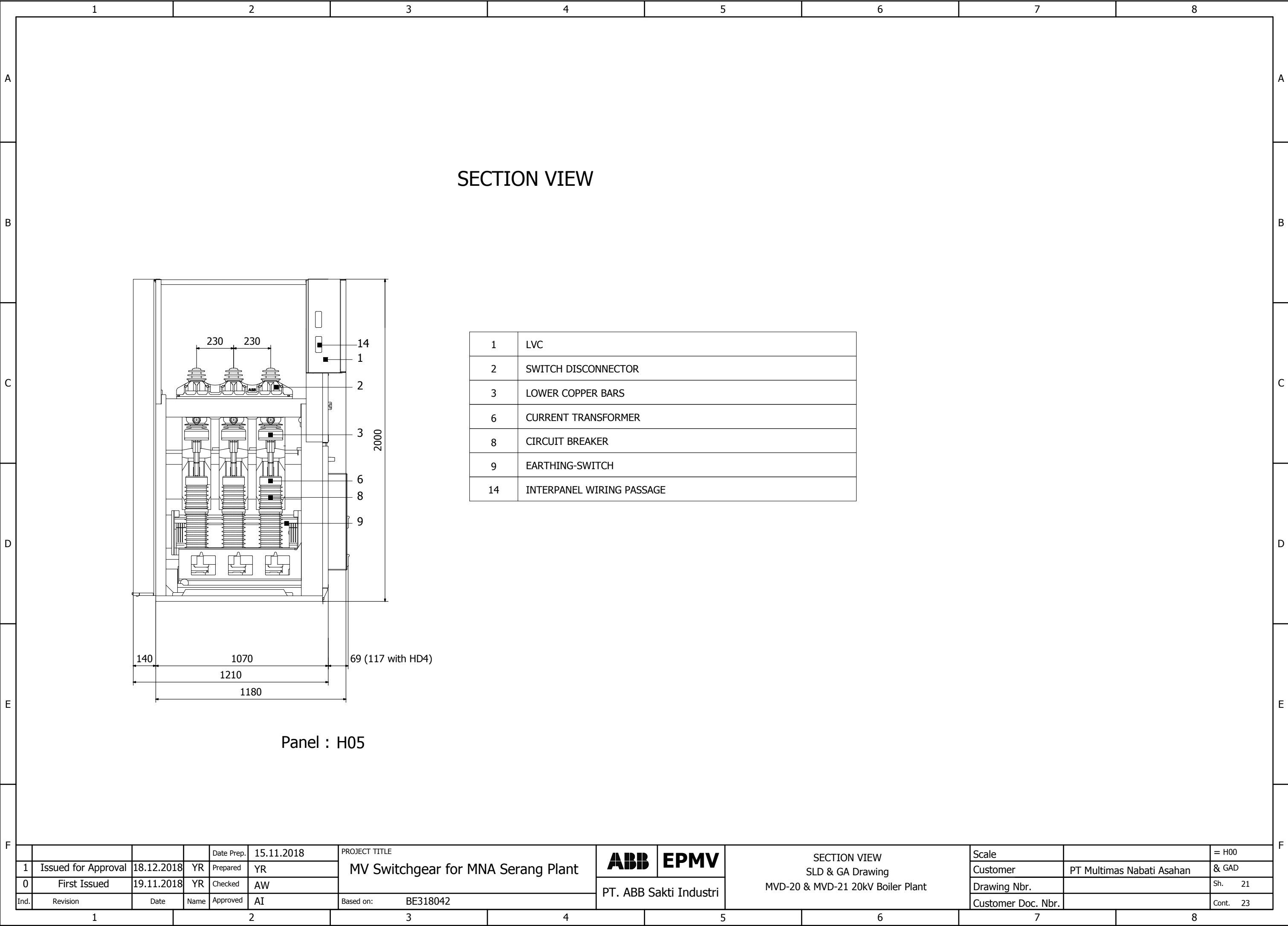
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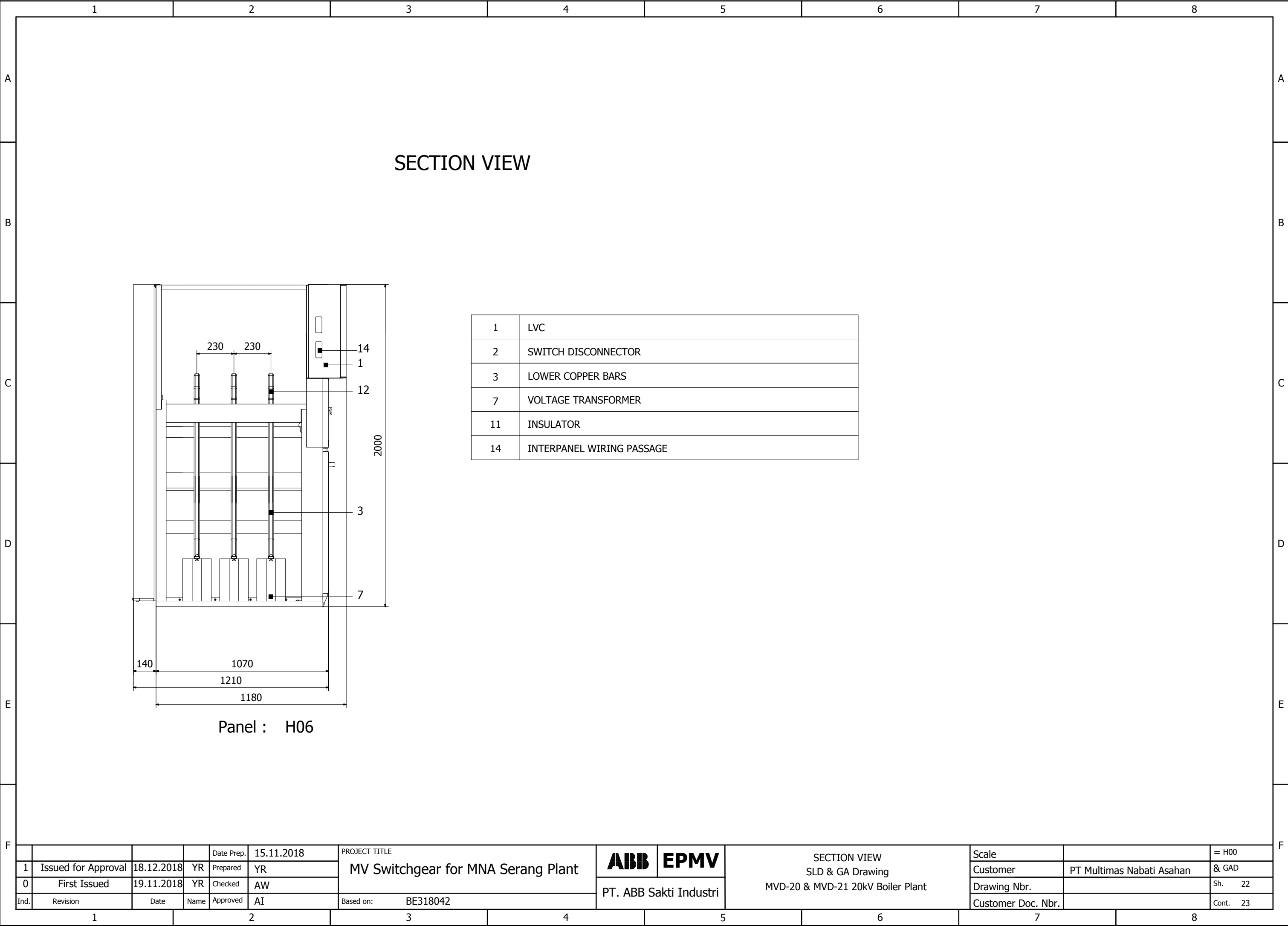


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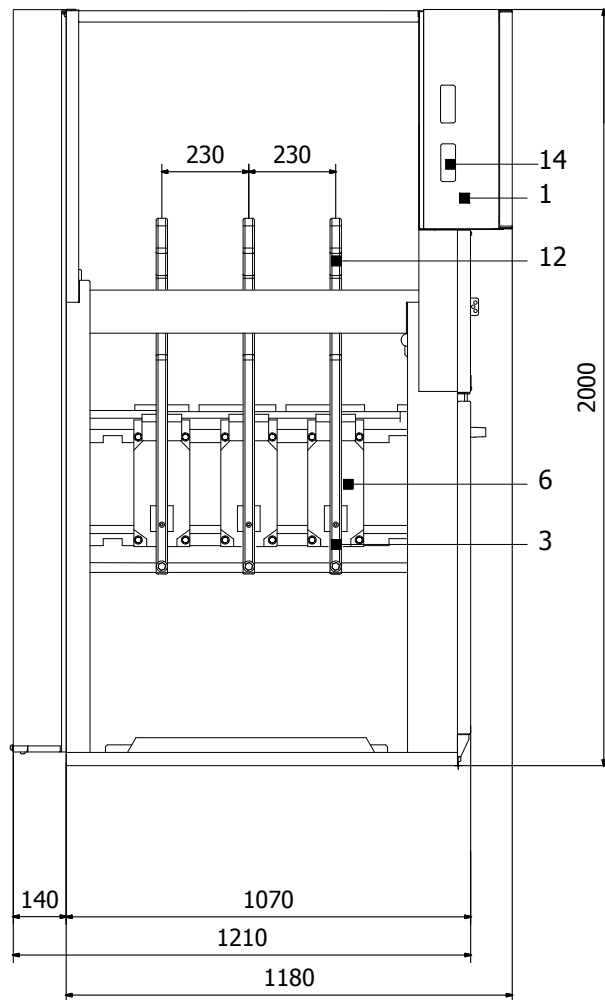


				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	SECTION VIEW SLD & GA Drawing MVD-20 & MVD-21 20kV Boiler Plant	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.			Sh. 21		
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SECTION VIEW



-	DISTANCE BETWEEN THE CABLE CONNECTION POINT AND THE FLOOR = 670mm
1	LVC
3	LOWER COPPER BARS
6	VOLTAGE TRANSFORMER
12	TOP COPPER BARS
14	INTERPANEL WIRING PASSAGE

Panel : H04

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	SECTION VIEW SLD & GA Drawing MVD-20 & MVD-21 20kV Boiler Plant	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					PT. ABB Sakti Industri	Customer	PT Multimas Nabati Asahan
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.				Sh. 23	
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